



User: Abdurrahman Kol Design: AG00976-BMC Otomotif_155Amp-ALT-AC-TM32 Secondary Drive 8PK1715HD E9843@-250 110_Corrected-IDR1-Posi 05June2025

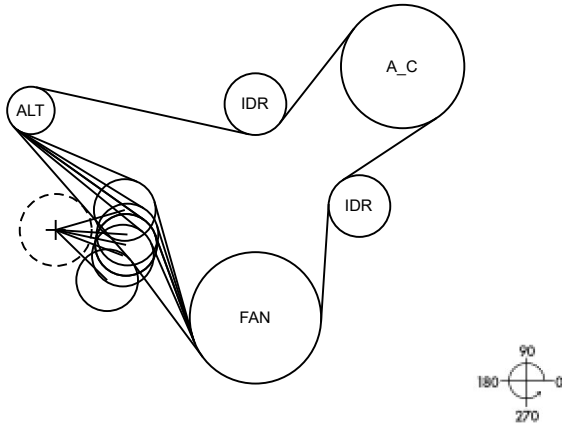
AG00976 BMC Otomotif FEAD 5
Cummins Eng.Scndr ALT&AC Drive
Ten. E9843 (22Nm); IDR E9839
Gates 8PK1715HD-Fleetrunner
Ten@-250/110; 05June2025

Belt Data, DB Ver. = 2.92.0.0, Belt Rev. = 7/14/2016

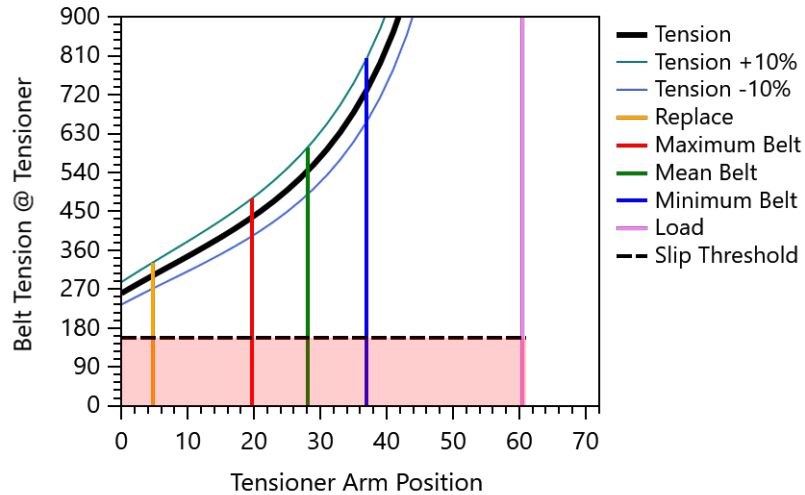
Belt Code	MT610
# of Ribs / Cord Material	8 / polyester
Effective Belt Length (ISO 9981)	1715

Tensioner Data

Type	Automatic
Design Tension	544 (122 lbf)



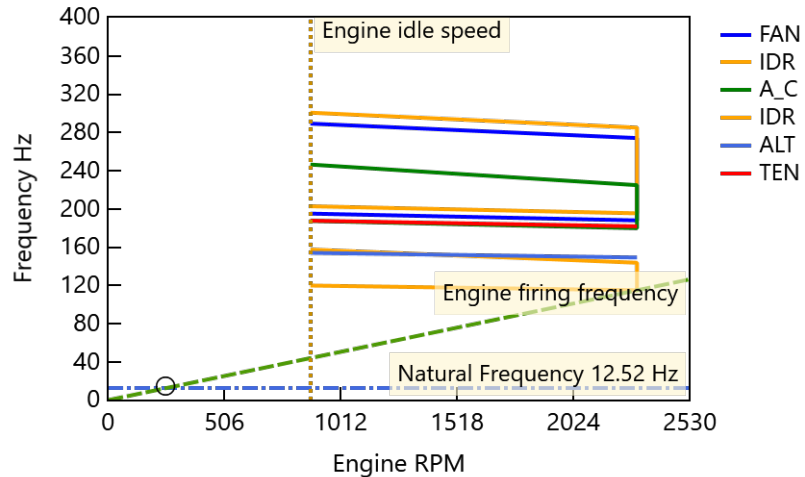
Belt Tension Control



Belt Life B10

Duty Cycle DCn-th %	Av. Vehicle Speed	PK-2_2p-MT3 hours
95	1	2670

Natural Frequency Map



Peak Tension & Hubload

	Engine RPM	Accel. RPM/s	Tension	Hubload	Direction /Wrap
FAN	880	1100	1585	2094.8	92 / 156
IDR	880	1100	1582	1409.1	330 / 53
A_C	880	1100	1177	2724.8	221 / 198
IDR	880	1100	1174	1251.6	110 / 64
ALT	880	1100	546	1690.3	340 / 157
TEN	880	1100	544	323.8	218 / 35

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)

Permissible Grooved Pulley

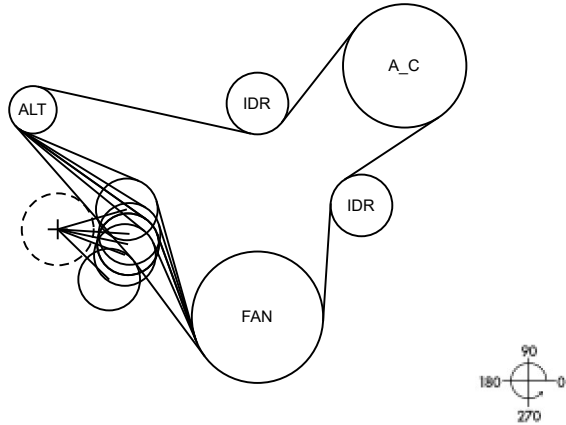
Axial Offset Allowance

For 0.90 Belt Entry Angle

Adjacent Grooved Pulleys	Flat Pulley Angular Misalignment	Axial Offset Allowance	Grooved Pulley Axial Offset %	Attributed to Flat Pulley Angular Misalignment %
A_C->FAN	IDR (0.33)	2.44	68	32
ALT->A_C	IDR (0.33)	2.23	48	52
FAN->ALT	TEN (1.20)	1.56	12	88



AG00976 BMC Otomotif FEAD 5
Cummins Eng.Scndr ALT&AC Drive
Ten. E9843 (22Nm); IDR E9839
Gates 8PK1715HD-Fleetrunner
Ten@-250/110; 05June2025



Layout Data

	X	Y	Flat	Pitch	Effective	DOB
FAN	0.00	0.00		164.40	162.00	162.99
IDR	130.10	139.90	75.00	77.20	79.60	
A_C	184.20	314.50		154.40	152.00	152.99
IDR	0.00	267.40	75.00	77.20	79.60	
ALT	-281.00	259.50		59.40	57.00	57.99
TEN	-161.97	91.29	75.00	77.20	79.60	

Belt Data, DB Ver. = 2.92.0.0, Belt Rev. = 7/14/2016

Belt Code	MT610
# of Ribs / Cord Material	8 / polyester
Stretch and Wear Allowance % of Length	0.60
Belt Length Tolerance (+/-)	6.00

Tensioner Data

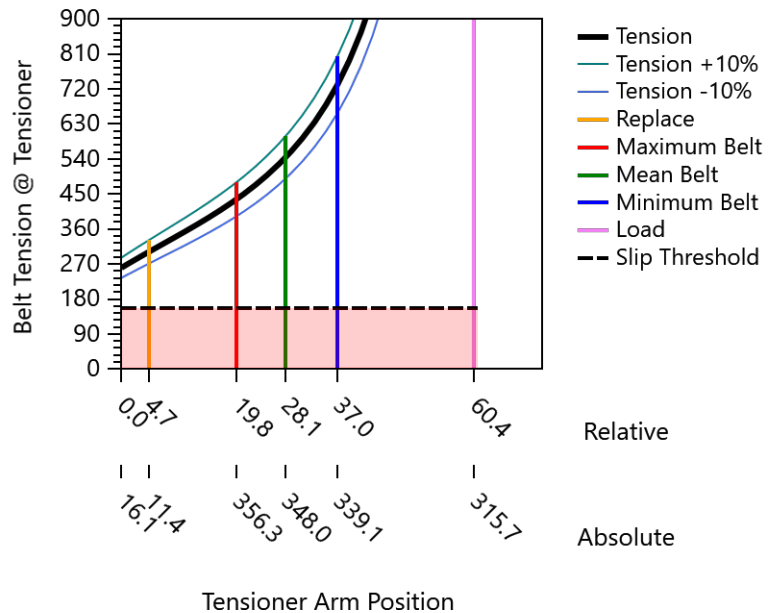
Type	Automatic
Design Tension	544 (122 lbf)
Pivot Point {X, Y Coordinates}	-250.00 , 110.00
Arm Length	90.0
Spring Mean Load	22.07
Spring Pre-Load	8.60
Spring Rate	0.480

RESULTS

Belt Drive System Geometry

	Span Length	Wrap Angle	Speed Ratio (Ref. Engine)
FAN	148.0	156.2	1.000
IDR	141.4	52.8	2.130
A_C	150.8	198.4	1.065
IDR	272.7	64.3	2.130
ALT	194.4	157.1	2.768
TEN	141.3	34.6	2.130

Belt Tension Control





AG00976 BMC Otomotif FEAD 5
Cummins Eng.Scndr ALT&AC Drive
Ten. E9843 (22Nm); IDR E9839
Gates 8PK1715HD-Fleetrunner
Ten@-250/110; 05June2025

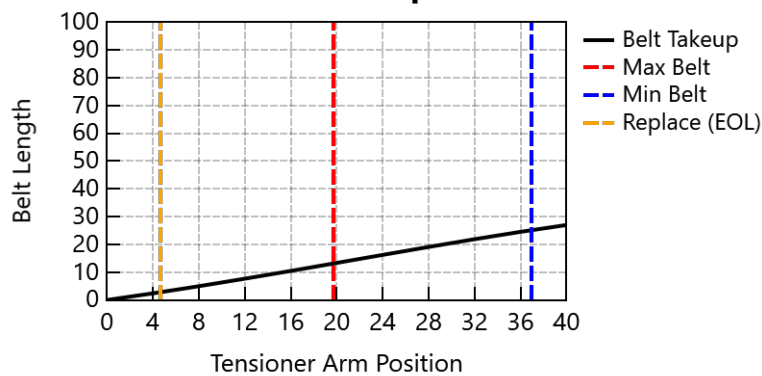
Tensioner Geometry

Belt Takeup / Tensioner Arm Ratio /

0.708

Position Description	Free Arm	Belt-Drive Length				Load
		Replace	Max	Mean	Min	
Arm Position	16.1	11.4	356.3	348.0	339.1	315.7
Idler X Coordinate	-163.5	-161.8	-160.2	-162.0	-165.9	-185.6
Idler Y Coordinate	134.9	127.8	104.3	91.3	77.9	47.1
Hubload to Arm Angle	25.2	28.9	41.6	49.4	58.2	83.3
Hubload Direction	221.3	220.3	218.0	217.4	217.3	218.9
Hubload	224.5	249.5	302.3	323.2	344.5	420.3
Tension	260.2	301.2	437.0	543.9	731.0	5222.8
Idler Wrap Angle	51.1	48.9	40.5	34.6	27.3	4.6
Effective Drive Length	1735.4	1732.5	1722.2	1716.2	1710.2	1700.7
Required Eff. Belt Length	1733.8	1730.9	1720.6	1714.6	1708.6	1699.1

Belt Take-up





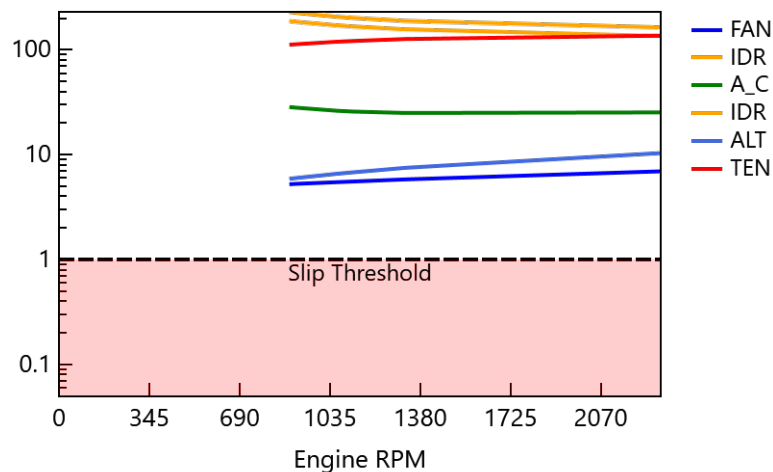
AG00976 BMC Otomotif FEAD 5
Cummins Eng.Scndr ALT&AC Drive
Ten. E9843 (22Nm); IDR E9839
Gates 8PK1715HD-Fleetrunner
Ten@-250/110; 05June2025

RESULTS

Belt Slip Sensitivity

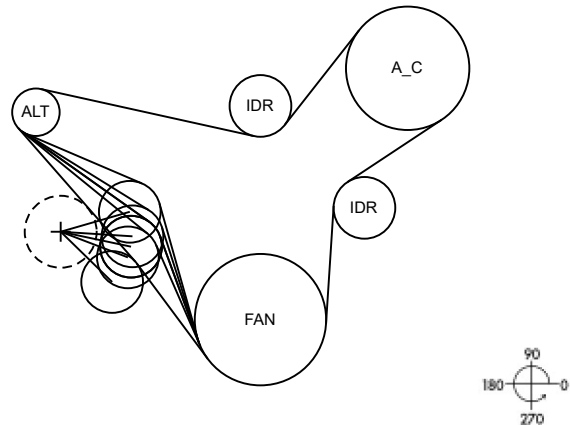
Pulley	Most Critical Load Condition						Required Increase Tension or Wrap Angle	
	Acceler. RPM/s	Load Combination %						
		A_C	ALT					
FAN	1100	100	100					
IDR	-1100	10	10					
A_C	-1100	100	10					
IDR	-1100	100	10					
ALT	1100	100	100					
TEN	1100	100	100					

Belt Slip Safety Factor





AG00976 BMC Otomotif FEAD 5
Cummins Eng.Scndr ALT&AC Drive
Ten. E9843 (22Nm); IDR E9839
Gates 8PK1715HD-Fleetrunner
Ten@-250/110; 05June2025



Load Conditions for DC 95%

%	Eng. RPM	V,	T,	FAN	IDR	A_C	IDR	ALT	TEN
25.00	880	1	90	6.34	0.01	2.70	0.01	3.61	0.01
4.00	1000	1	90	7.41	0.01	3.60	0.01	3.78	0.01
4.50	1100	1	90	7.86	0.01	4.00	0.01	3.83	0.01
5.00	1200	1	90	8.30	0.01	4.40	0.01	3.87	0.01
5.50	1400	1	90	9.06	0.01	5.10	0.01	3.93	0.01
6.00	1500	1	90	9.37	0.01	5.40	0.01	3.94	0.01
7.50	1600	1	90	9.42	0.01	5.70	0.01	3.69	0.01
9.00	1700	1	90	10.00	0.01	6.00	0.01	3.97	0.01
0.50	1800	1	90	10.31	0.01	6.30	0.01	3.98	0.01
12.00	2000	1	90	10.72	0.01	6.70	0.01	3.99	0.01
16.00	2500	1	90	11.03	0.01	7.00	0.01	4.00	0.01
5.00	2750	1	90	11.45	0.01	7.40	0.01	4.02	0.01

Mean Hubloads for DC 95%

%	Eng RPM	FAN	IDR	A_C	IDR	ALT	TEN
25.00	880	1891	1228	2372	1088	1539	324
4.00	1000	1915	1249	2359	1048	1501	324
4.50	1100	1885	1222	2295	1011	1467	324
5.00	1200	1858	1198	2239	979	1437	324
5.50	1400	1807	1152	2140	927	1389	323
6.00	1500	1781	1129	2093	905	1368	323
7.50	1600	1740	1092	2015	865	1331	323
9.00	1700	1739	1092	2018	869	1334	323
0.50	1800	1721	1076	1986	853	1320	323
12.00	2000	1679	1038	1919	827	1295	323
16.00	2500	1570	940	1765	778	1250	323
5.00	2750	1541	914	1720	760	1234	323

Mean Tensions for DC 95%

%	Eng RPM	FAN	IDR	A_C	IDR	ALT	TEN
25.00	880	1381	1380	1023	1022	545	544
4.00	1000	1405	1404	985	984	545	544
4.50	1100	1374	1373	951	949	545	544
5.00	1200	1347	1346	921	920	545	544
5.50	1400	1296	1295	872	871	545	544
6.00	1500	1270	1269	851	850	545	544
7.50	1600	1228	1227	813	813	545	544
9.00	1700	1227	1227	817	816	545	544
0.50	1800	1209	1209	802	801	545	544
12.00	2000	1167	1166	777	776	545	544
16.00	2500	1056	1056	731	730	544	544
5.00	2750	1028	1027	715	714	544	544